# Omar Ahmad Elhakim

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## Education

**Ain Shams University** **Expected Graduation June 2026**
Senior Computer Science Student, Scientific Computing Department | GPA: 3.3

## Experience

**Zinad** **July 2024 – Present**
*AI Engineer (Part-Time)*

- Engineered an AI-powered email analysis pipeline using Python, LangChain, and advanced prompt engineering to detect phishing indicators such as tone, context, and grammatical anomalies.
- Developed automated generative workflows using LLMs to dynamically construct context-aware phishing simulation templates and realistic landing pages from scratch based on user parameters.
- Integrated real-time AI conversational agents into the main application, optimizing prompt latency and response streaming interactions.
- Collaborated on refactoring AI service calls into a robust backend architecture to handle asynchronous processing efficiently.

## Graduation Project

**EEG-Based Emotion Classification (Neuromarketing)**

- Built a data preprocessing pipeline in Python to clean and extract Differential Entropy (DE) and spatial features from the SEED-IV EEG dataset.
- Developed and trained deep learning models in PyTorch, specifically Directed Graph CNNs (DGCNN), Continuous CNNs (CCNN) and other transformer-based models to classify distinct emotional states.
- Fine-tuned hyperparameters to achieve over 84% accuracy on 4-class emotion recognition using DGCNN.

## Technical Projects

**Machinery Fault Diagnosis (Vibration Analysis)**

- Developed a time-series classification model to detect mechanical faults using PyTorch, Fastai, and Tsai, leveraging Optuna for hyperparameter tuning to optimize the F1 score.

**Food & Fruit Recognition**

- Trained a robust computer vision model in Python to classify various types of food and fruits, applying dataset augmentation, feature extraction, and extensive performance evaluation.

**Indoor Style Image Classification**

- Built a machine learning model to categorize architectural indoor styles from images, managing the end-to-end process from raw dataset preprocessing to final model inference.

## Technical Skills

**Languages:** Python, SQL, C/C++, Java, PHP, Bash
**Tools/Frameworks:** PyTorch, LangChain, Pandas, NumPy, Optuna, Git, Github, Linux, Docker, Laravel

## Professional Development

- **MEC Model of Leadership:** Completed an extensive leadership program (160 hours) covering emotional intelligence, effective communication, self-awareness (50 hours), the Enneagram, problem-solving, and team building.